# Safety

*Claude edition — the boundary comes first*

Bound the task, rehearse it cold, then collect evidence

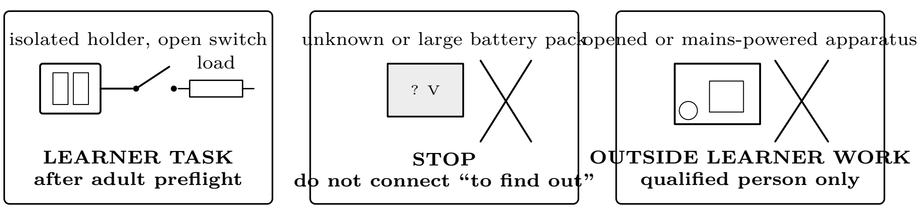
**LEARNER LIMIT**

Learner investigations use only isolated battery sources at 12 VDC or less. Outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, and exposed high voltage are not learner work. An adult facilitator controls the bounded battery activity; adult status does not make anyone electrically qualified.

**Three roles—never collapse them**

| | |
|---|---|
| **Learner** | Predicts, assembles only the approved battery circuit while deenergized, observes, records, and obeys every stop call. |
| **Adult facilitator** | Selects materials, verifies the boundary and meter setup, owns the switch and timing decision, and ends the trial at uncertainty. |
| **Qualified person** | Performs work whose voltage, equipment, environment, or rules require electrical qualification. Being the learner's parent, teacher, or supervising adult does not confer this role. |

## Sort the task before touching parts




**Boundary questions**

1. The label says 9 V, but the pack is large and its current capability is unknown. Which part of the boundary rejects it?
2. An appliance is unplugged. Does that remove stored energy, damaged insulation, or the need for qualification?
3. An adult offers to make the connection. Does changing the hands performing the work change the work's category?

## Preflight: point, name, verify

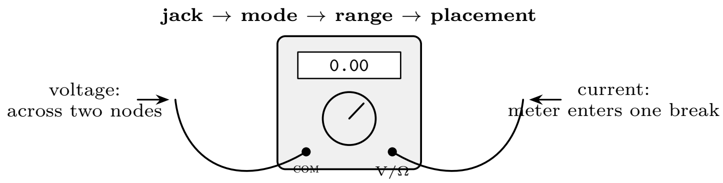
| Gate | Learner points and says | Adult verifies |
|---|---|---|
| Source | Type, cell count, stated voltage | Isolated holder; correct intact cells; no swelling, leaks, dents, corrosion, heat, or mixed chemistries |
| Workspace | Switch, disconnect, parts tray | Dry, bright, clear of drinks, jewelry, loose metal, and bystanders; eye protection in place |
| Conductors | Every intended bare end | Insulation sound; strands contained; no possible bridge across source terminals |
| Load | Part name and expected behavior | Correct value/rating; LED has its series resistor; coil or resistor shows no prior heat damage |
| Control | Who owns the switch; trial duration | Switch begins open; disconnect reachable; timer and cool-down interval ready |
| Evidence | Predicted sign, direction, or range | Prediction is specific enough to distinguish expected, null, and surprising results |

**DECISION**

If a label, rating, connection, meter setting, or expected result is uncertain, the source remains disconnected. “Probably” is a failed preflight.

**Checkpoint.** One copper strand nearly reaches the other terminal. It has not touched. Explain why this is already a failed conductor gate.

## Meter rehearsal—battery holder empty




1. Inspect the case, lead insulation, and probe guards. Remove a damaged meter or lead from service; tape is not a bench repair.
2. Put black in COM. Put red in the jack named by the lesson. Select the quantity and a range that can contain the predicted value.
3. Rehearse voltage by touching the two named nodes with the empty holder. Rehearse current by opening the named path and inserting the meter there.
4. Never place a current-configured meter across a source. Return the red lead from a current jack when the current measurement ends.
5. Adult introduces one cold fault: wrong jack, wrong mode, open connection, or reversed LED. Learner finds it by inspection, never by trial power.

**Intermediate meter questions**

The display remains zero. Could that mean a true zero, open switch, loose connection, wrong range, or wrong placement? What must happen before moving a lead? **Open the switch and disconnect.**
The expected current is 18 mA. The probes are laid across the holder. Why must the adult stop the trial even though the source is within the voltage limit?

## One controlled energizing cycle

1. Adult completes the preflight and calls “ready.” Learner’s hands clear the circuit. The switch owner calls “on” and starts the stated interval.
2. Observe only the planned quantity. Never reposition, squeeze, substitute, or add a part while energized.
3. Switch open; source disconnected; adult checks for unexpected heating. Only then may a lead, probe, load, core, or magnet move.
4. Record the result with units and trial time. A null or surprising result begins a deenergized diagnosis; it never authorizes more voltage.

**STOP MEANS THE TRIAL IS OVER**

Stop for unexpected warmth, a hot cell or lead, swelling, leakage, sharp odor, smoke, discoloration, buzzing, sparking, pain or tingling, damaged insulation, a dropped metal object, unstable readings, or behavior outside the prediction. Open the switch if safe, disconnect without touching the suspect part, isolate the setup, and tell the adult. Do not repeat “just once.”

## Fault decisions

| Observation | Immediate action | Cold diagnosis |
|---|---|---|
| No response | Open, disconnect, preserve layout | Compare each node with the drawing; check continuity and polarity; do not raise source voltage |
| Meter overload or jumping digits | Disconnect before touching probes | Recheck jack, mode, range, placement, and predicted magnitude |
| Part warms more each trial | Stop and leave it untouched | Check value, wiring, current, duration, and cool-down; replace damaged parts |
| Cell heats, leaks, smells, dents, or swells | Adult clears and isolates the area | Follow cell-maker and local disposal guidance; never reuse or recharge it |
| Metal bridges source terminals | Disconnect if safe; never grab hot metal | Replace damaged holder, switch, cells, or leads; rebuild from the start |
| Small magnet or clip is missing | Stop and count all pieces | Find it before anyone leaves; store magnets inaccessible to children |

**Falsification questions**

1. What evidence would show that an apparent “bad resistor” is actually an open switch or lead?
2. Three timed coil trials lift fewer clips each time. Name a thermal change that could matter even though turns and source stayed the same.
3. Why does an unexpected result count against the setup, not against the safety boundary?

## Cold verification drill

Use an approved lesson circuit with the holder empty. Learner speaks and points; adult evaluates without prompting.

1. State source type and voltage; identify switch, load, and disconnect.
2. Trace the intended current path and the most plausible accidental short.
3. Demonstrate both meter placements and restore the red lead to V/Ω afterward.
4. Name five stop observations and show where hands go when “stop” is called.
5. Find an adult-created harmless fault by working through the gates in order. Do not insert cells to diagnose it.

**VERIFICATION—EVERY BOX BEFORE CELLS ENTER**

☐ Isolated source ☐ $\leq$ 12 V DC ☐ Correct parts ☐ Intact cells/leads ☐ Switch open ☐ Shorts removed ☐ Meter verified ☐ Prediction written ☐ Time stated ☐ Stop signs named

**Final evidence record**

Boundary decision ____________________

Fault found cold ____________________

Meter rehearsal ____________________

Stop response ____________________

Adult verification ____________________

Safety basis: OSHA 29 CFR 1910.333 and OSHA Publication 3075; NIST SI references; U.S. Department of Energy electrical safety handbooks. Full citations and scope notes are recorded in the provider-neutral electricity research library.